

1st Progress Report

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Date:	16/1/2024
Subject:	RL for Flying-V v2.0
Supervisors:	Dr. E.J. van Kampen
Project term:	01/2024 - 08/2024
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Outcome of the previous meeting

- Decided on RL for Flying-V 🎉.
- Mentioned various forms that the project can take (symmetric vs asymmetric model/controller, offline vs online vs hybrid or different RL algorithms). Specific problem & approach not defined yet, kept open for now.
- Submitted application for knowledge embargo, submitted kickoff form to SPA-LR.
- Briefly mentioned planning of thesis, length of literature study phase, ($\approx$ 2-3 months).

Description of the results since last meeting¹

- Read Willems' thesis² in detail, i will bootstrap 😊 my literature review with knowledge gleamed from this thesis.
- Listed theses done on the topic of RL flight control under Erik-Jan's supervision, summarized each thesis.
- Drafted a gantt chart and allocated approximate time periods for the major phases, and for first half of literature study.
- Began reading Sutton & Barto's Reinforcement learning textbook.

Plans for next 2 weeks

All the items enumerated below are to lead up to writing the first iteration of research objective & questions, planned to be done in week 4³.

1. Wrap up reading on Sutton & Barto (finish Part-I detailed reading, do skimmed reading on Part-II)
2. Gather and study literature on flying-V.
3. Survey state-of-the-art RL algorithms (special interest on hierarchical RL, different hybrid algorithm types, evolutionary RL)
4. If spare time available, study implementing RL algorithms.

¹all my notes can be found here: https://github.com/wingos80/thesis_notes.git

²just to be clear: <http://resolver.tudelft.nl/uuid:a6b645d2-8d47-44d3-a4ad-1d5a6024f13f>

³week of 29th Jan to 4th Feb, right before when i go for spring break from 4th Feb to 9th Feb.